



Impact of In-Situ Functionalization of Carbon Nanotubes Using Bis(Triethoxysilylpropyl) Tetrasulfide on the Properties of EPDM/SBR–CNT Composites

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Abstract

Carbon nanotube (CNT) composites were developed by blending ethylene propylene diene monomer (EPDM) and styrene butadiene rubber (SBR) with in-situ functionalized CNTs using the silane coupling agent bis(triethoxysilylpropyl) tetrasulfide (TESPT). This study examines the impact of TESPT-functionalized CNTs (tCNTs) on the properties of vulcanized EPDM/SBR, focusing on rheological behavior, mechanical performance, abrasion resistance, compression set, crosslinking density, and swelling resistance. Rheometric analysis indicated that increasing tCNT content results in reduced scorch time and optimum cure time, while minimum torque, maximum torque, delta torque, and the cure rate index increase, suggesting stronger polymer-filler interactions. The stress at 100% elongation and tensile strength significantly improved up to 5 phr tCNT, with further additions causing diminishing returns due to increased filler-filler interactions. Moreover, tear strength, hardness, abrasion resistance, compression set, crosslinking density, and swelling resistance all improved with rising tCNT content, demonstrating the strong reinforcing capabilities of tCNTs in enhancing the mechanical and physical properties of EPDM/SBR composites.

Keywords tCNT · EPDM/SBR blend · Rheometric behavior · Mechanical performance

1 Introduction

Styrene-butadiene rubber (SBR) and ethylene propylene diene monomer (EPDM) are two extensively used synthetic rubbers, known for their distinct characteristics [1–3].

SBR is valued for its excellent abrasion resistance, aging stability, and high resilience, making it ideal for applications such as tires, footwear, and conveyor belts. However, its limitations include poor resistance to weather, ozone, and chemicals, particularly in outdoor environments [4–6]. Conversely, EPDM exhibits excellent resistance to UV light, ozone, weathering, and heat, making it ideal for outdoor applications such as roofing membranes, automotive seals, and hoses. However, EPDM has lower resistance to oils and fuels, restricting its use in certain industrial applications [7–9]. While SBR offers better wear resistance and cost-effectiveness, EPDM provides superior environmental durability. Blending the two rubbers achieves a balance of properties, making them suitable for various applications like automotive components, weather seals, and vibration-dampening systems, where both durability and flexibility are crucial [10–15].

Furthermore, the incorporation of fillers like calcium carbonate (CaCO₃), silica (SiO₂), carbon black (CB), clay, carbon nanotubes (CNTs), and carbon fibers (CFs) [16–23] can enhance rubber materials' properties, including strength,

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modulus, stiffness, and dynamic performance. CNTs, in particular, provide a unique combination of electrical and thermal conductivity, along with exceptional modulus and strength, positioning them as highly promising fillers for rubber composites [24–27]. As a result, the combination of CNTs with EPDM/SBR blends has garnered significant interest due to the potential for enhanced composite properties.

To improve the performance of polymer materials, the use of novel fillers and advancements in processing technologies have become common practice [28]. Nanocomposites, materials that incorporate fillers on the nanometer scale, have demonstrated considerable potential for improving the properties of polymers [29–31]. Because nanoparticles have a larger surface area and surface energy than conventional fillers, research has demonstrated that their addition can significantly improve polymer performance. Since the discovery of CNTs in 1991, their outstanding physical and mechanical properties have been the focus of extensive studies [32–35]. With advancements in CNT synthesis, their batch production has become feasible, and CNTs are anticipated to be widely used as fillers in polymers to enhance physic-mechanical properties.

Although CNTs have been extensively studied in various polymers, their application in rubber materials remains relatively underexplored [36–39]. Rubber materials' great and reversible deformability makes them essential for industrial applications. However, reinforcement using fillers is often required since their inherent modulus and durability often fall short of practical requirements [40–42]. Traditionally, CB or SiO₂ is used to enhance rubber performance, but there remains potential for further improvement. Incorporating nanoscale fillers like CNTs has proven more effective than microscale fillers, as their higher specific surface area enhances rubber properties while preserving the material's elasticity. Further exploration of CNT-reinforced rubber nanocomposites presents significant opportunities for both industrial and societal advancements [43–44].

By incorporating clay into rubber matrices, the idea of nanoscale filler-reinforced rubber was initially introduced. The modulus of natural rubber increased significantly when 10 weight% layered silicates were added. Because of their better qualities, CNTs are expected to offer even more reinforcement. Despite the fact that CNT-reinforced rubber has been the subject of numerous research, the high aspect ratio of the CNTs and the high viscosity of the rubber make it difficult to achieve a uniform dispersion of CNTs throughout the rubber. Furthermore, there is still a lack of ideal interaction between the CNTs and the rubber matrix. Consequently, the performance of CNT-reinforced rubber nanocomposites has not reached its full potential, emphasizing the necessity for further development to enhance these materials [45–46].

It has been shown that silane coupling agents are effective in increasing the dispersion of CNTs in rubber matrices by creating chemical interactions between the CNT surfaces and rubber molecules. Prior research on carbon black and ex-situ functionalised carbon nanotubes (CNTs) utilising bis(triethoxysilylpropyl) tetrasulfide (TESPT) in natural rubber (NR) composites showed that these composites had better mechanical and dynamic properties than those with non-functionalized CNTs. The chemical interactions between the NR molecules and the silanized CNTs were credited with this improvement. However, the actual applicability of ex-situ functionalisation is limited due to its time-consuming and expensive nature. As a result, in NR–CNT composites, researchers have already functionalised CNTs with TESPT [47–48].

EPDM/SBR–CNT composites offer improved tensile strength, flexibility, and durability, making them ideal for automotive seals, hoses, and industrial rubber products requiring enhanced mechanical properties and resistance to wear. In this study, EPDM/SBR–CNT composites were developed through the in-situ functionalization of CNTs with silanes. This method entails the direct incorporation of silanes into the open-mill mixer during the compounding process, thereby negating the necessity for a wet chemical pretreatment of the CNTs. The primary goal of this approach is to enhance the dispersion of CNTs within the EPDM/SBR matrix, which is the main focus of the functionalization. Furthermore, the research investigated the impact of in-situ functionalization on the processing characteristics and properties of the EPDM/SBR–CNT composites.

2 Experimental Part

2.1 Materials

EPDM KEP270 [49] and SBR-1502 [50] were acquired from Arihant Reclamation Private Limited, Delhi, India. The multi-walled carbon nanotubes (MWCNTs) [51] utilized in this study were provided by Shenzhen Nano Technologies Port Co. Ltd. in China. Bis(triethoxysilylpropyl) tetrasulfide (TESPT) and other rubber chemicals were sourced from Vignesh Chemicals, while solvents were acquired from Sigma-Aldrich through Subra Chemicals, Puducherry, India, and were used without further purification.

2.2 Preparation of EPDM/SBR–CNT Composites

Melt blending in an open-mill mixer was used to create the EPDM/SBR–CNT composites. First, raw CNTs and the EPDM/SBR blend were combined at 60 °C for 6 min to create silane-free composites. Other chemicals like sulphur,

zinc oxide, stearic acid, and an accelerator were added after this first mixing. Raw CNTs and the EPDM/SBR mixture were combined for five minutes at temperatures between 70 and 90 °C for the composites that underwent in-situ functionalisation. In order to maintain a silane concentration of 0.06 ml/g of CNT, the silane coupling agent (SCA or TESPT) was then introduced straight to the open-mill mixer and mixed for a further five minutes. After cooling the mixture to room temperature, the open-mill mixer was used to add the remaining chemicals one at a time at 50 °C. For all composite types, a second mixing phase was carried out using a two-roll mill to improve the dispersion of CNTs inside the rubber matrix and to sheet the samples. Table 1 provides the detailed compounding formulations for the nanocomposites, presenting a comprehensive breakdown of the proportions of each material in the mixture. Finally, composite sheets measuring 150×160×2 mm³ were produced through compression molding at 160 °C, with the molding duration determined by the results of rheometer tests.

2.3 Testing and Characterization

Cure characteristics: An oscillating disc rheometer (ODR 2000, Monsanto Co. Ltd., Ohio, USA) was used to evaluate the curing properties of virgin EPDM/SBR and its CNT composites. Setting the rheometer to run at 1.67 Hz and 1 arc degree of amplitude while maintaining a constant temperature of 160 °C throughout the experiments was part of the testing process. The samples were prepared and placed into the rheometer chamber, where the oscillatory shear conditions facilitated the evaluation of the viscoelastic properties and curing behavior of the EPDM/SBR compounds [52–53]. The information acquired during this measurement allowed for a thorough examination of a number

of crucial cure characteristics, such as the cure rate index (CRI), maximum torque (M_H), minimum torque (M_L), delta torque (ΔM), optimum cure time (t_{90}), and scorch time (t_{s2}).

Tensile and tear properties: A tensile testing machine from Dak System Inc. was used to perform tensile and tear tests at a regulated temperature of 23 ± 2 °C. Dumbbell-shaped specimens were made for the tensile testing in accordance with ASTM D-412 standards, guaranteeing uniform measurements and the right amount of material thickness [54–56]. Similarly, tear samples were prepared in accordance with ASTM D-624 to facilitate accurate measurement of tear strength [57–58]. During testing, the machine operated at a cross-head speed of 500 mm/min, allowing for a gradual application of force until failure occurred. The results provided essential information about the mechanical properties of the materials, including tensile strength (the maximum tensile stress a material can endure before failing [59]), elongation at break (the percentage increase in length of the material when stretched to the point of breaking, indicating its ductility [60]), stress at 100% elongation (the stress level the material experiences when stretched to twice its original length [61]), and tear strength (the material's resistance to tear propagation, measured as the force required to continue tearing it [62]). For each sample, the values reported are averages based on five measurements.

Hardness: The assessment of hardness was performed using a Shore-A durometer, as outlined in ASTM D-2240. Prior to testing, the samples were prepared to ensure a flat, uniform surface, free from imperfections. The Shore-A durometer was calibrated according to the manufacturer's specifications to ensure accuracy. The durometer's indenter was then placed firmly on the surface of the specimen, applying a consistent pressure for a specified duration to allow for proper indentation [63]. Measurements were taken at various points on each sample to ensure precision. The

Table 1 Formulation for EPDM/SBR/tCNTs rubber blends

Sample ID	Ingredients (phr ^a)							
	EPDM ^b	SBR ^c	tCNTs ^c	Zinc oxide	Stearic acid	MBTS ^d	TMTD ^e	Sulphur
tCNTs ₀	80	20	0	4	1.5	1.2	1	2.5
tCNTs ₁	80	20	1	4	1.5	1.2	1	2.5
tCNTs ₂	80	20	2	4	1.5	1.2	1	2.5
tCNTs ₃	80	20	3	4	1.5	1.2	1	2.5
tCNTs ₄	80	20	4	4	1.5	1.2	1	2.5
tCNTs ₅	80	20	5	4	1.5	1.2	1	2.5
tCNTs ₆	80	20	6	4	1.5	1.2	1	2.5
tCNTs ₇	80	20	7	4	1.5	1.2	1	2.5
tCNTs ₈	80	20	8	4	1.5	1.2	1	2.5

^a (phr): parts per hundred rubber

^b(EPDM): ethylene propylene diene monomer

^c (SBR): styrene butadiene rubber

^d(MBTS): mercaptobenzothiazole disulfide

^e (TMTD): tetramethylthiuram disulfide

hardness value reported was the most consistent and repeatable result obtained during the testing process.

Rebound resilience: To evaluate rebound resilience, the elastomeric nanocomposites were tested according to ASTM D 2632. The test involved suspending the specimen at a specified height above the plunger of a vertical rebound tester (resiliometer). The plunger was then released, and the rebound height of the specimen was measured. The procedure was conducted five times for each sample, and the average rebound height was noted to guarantee consistent and accurate results [64]. To determine the rebound resilience—an indicator of the material's ability to recover after deformation—the ratio of the rebound height to the initial height was calculated, with the average of the five measurements reported for each sample.

Abrasion resistance: The abrasion resistance of the rubber nanocomposites was assessed using a DIN abrader, specifically the Zwick Abrasion Tester model 6102, in accordance with the standards ASTM D5963, DIN 53,516, and ISO 4649. For testing purposes, cylindrical specimens were prepared with a diameter of 10 ± 0.2 mm and a thickness of 12 ± 0.2 mm. These specimens were abraded on grade 60 emery paper at a speed of 0.32 m/s while applying a force of 10 N. The test was carried out over a total distance of 40 m. After the abrasion test, the volume loss of the specimens was determined [65]. The average abrasion loss for each sample was calculated based on at least five specimens using Eq. (1), ensuring accurate and consistent measurement of wear resistance.

$$\text{Abrasion loss (mm}^3\text{)} = (\Delta m \times S_0)/(\rho \times S) \quad (1)$$

In this context, ρ represents the material density in milligrams per cubic millimeter (mg/mm^3), Δm corresponds to the mass loss of the specimen in milligrams (mg) due to abrasion, S denotes the average abrasive power measured during the test in milligrams (mg), and S_0 is the nominal abrasive power, which is set at 200 mg as a standard reference value.

Swelling resistance: A sample measuring $25 \times 25 \times 2$ mm³ was cut from the vulcanized rubber sheet and dried for 24 h to ensure it was free of moisture. The thickness of the sample was subsequently measured with a Vernier caliper, which has an accuracy of ± 0.01 mm. The prepared sample was then subjected to the ASTM D471 swelling test. Initially, the sample was weighed on an electronic scale that provides an accuracy of ± 0.001 g. The sample was immersed in glass diffusion bottles filled with various hydrocarbons. At set intervals, the sample was taken out, cleaned to remove any surface contaminants, and then reweighed. This process continued until equilibrium swelling was achieved [66]. The mole percent uptake (MPU), denoted as Q_t , was calculated

using the appropriate formula to determine the degree of swelling in the hydrocarbons.

$$Q_t (\text{mol}\%) = \frac{(M_t - M_0)/\text{Molecular weight of the solvent}}{M_0} \times 100 \quad (2)$$

In this context, M_t represents the mass of the specimen measured after 72 h of immersion in the hydrocarbons, while M_0 denotes the initial mass of the specimen before immersion.

Compression set: The compression set test was performed following ASTM D395 standards. The rectangular plates of the compression apparatus were precisely aligned around the test sample, using spacers for proper positioning. The sample was then secured and compressed to 25% of its original thickness. Following this, the assembly was placed in a 70 °C oven for 24 h to allow for the effects of compression to take place. After the heating period, the sample was removed and allowed to cool for 30 min [66]. The final thickness of the specimen was subsequently measured using an electronic digital Vernier caliper, ensuring an accuracy of 0.01 mm. A specific formula was applied to calculate the compression set, which quantifies the level of deformation retained by the material after the compression process.

$$\text{Percentage of compression set, } C\% = \frac{t_0 - t_1}{t_0} \times 100 \quad (3)$$

In this context, t_0 represents the initial thickness of the specimen before compression, while t_1 denotes the thickness of the specimen measured after it has been removed from the compression device. Additionally, t_s refers to the thickness of the spacer bar used during the compression process.

Morphology: The fractured surface morphology of the tensile test samples was examined using a field emission scanning electron microscope (FE-SEM) from Hitachi (model S-3400). Before the analysis, the samples were coated with a thin layer of gold to enhance conductivity and improve image quality. The FE-SEM was operated at an accelerating voltage of 3 kV to capture high-resolution images of the fracture surfaces.

Crosslinking Density: To assess the cross-link densities, rectangular test pieces of pristine EPDM/SBR and its composites, each measuring $4 \times 2 \times 25$ mm³, were prepared using the equilibrium swelling method. The samples were immersed in toluene for 48 h to provide adequate time for the solvent to permeate the polymer matrix. After immersion, the swollen specimens were removed, and any excess solvent on their surfaces was carefully wiped away. The cross-link densities were subsequently determined using the Flory–Rehner equation, as described in Eq. (4) [67].

$$M_c = \frac{-\rho_p V_s V_r^{1/3}}{\ln(1 - V_r) + V_r + \chi V_r^2} \quad (4)$$

In this context, M_c represents the molar mass of the EPDM/SBR polymer between crosslinks, while V_s denotes the molar volume of toluene, the solvent used, which is 106.3 mL/gmol. Additionally, ρ_p refers to the density of the EPDM/SBR rubber blend. The volume fraction of the polymer in the solvent-swollen compound is indicated by V_r , and the solvent-polymer interaction parameter, denoted as χ , is given a value of 0.3. The volume fraction V_r can be determined using the specific equation provided in the referenced literature.

$$V_r = \frac{1}{1 + Q_m} \quad (5)$$

In this context, Q_m represents the weight swell of the EPDM/SBR-CNTs nanocomposites when immersed in toluene. The degree of cross-link density, denoted as v , is calculated using the equation presented in Eq. (6).

$$v = \frac{1}{2M_c} \quad (6)$$

3 Results and Discussions

Cure characteristics refer to the various parameters that define the curing process of rubber materials, including t_{s2} , t_{90} , M_L , M_H , ΔM , and CRI. t_{s2} indicates the stability of the compound during processing and is the time taken before the onset of vulcanization. t_{90} is the period needed for a rubber compound to reach its best vulcanization level, ensuring peak properties without risking degradation. M_L represents the lowest viscosity measured during the curing process, reflecting the compound's processability, while M_H indicates the highest viscosity observed, which reveals the maximum elasticity of the material. ΔM , the difference between M_H and M_L , provides insight into the overall cure characteristics and filler interactions, and the CRI measures the speed of vulcanization, indicating how quickly the material reaches its optimum properties.

In Table 2, the M_L of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. As the concentration of tCNTs rises, a significant increase in the M_L of the tCNT-reinforced EPDM/SBR nanocomposites is observed. This improvement is primarily due to enhanced interactions between the tCNTs and the matrix (EPDM/SBR blend), as tCNTs promote a more effective dispersion of CNTs within the EPDM/SBR matrix [68]. As a result, there is a more efficient transfer of stress between the tCNTs and the EPDM/SBR, which reinforces these interactions and restricts the mobility of the EPDM/SBR (polymer) chains. This leads to greater stiffness and enhanced resistance to

Table 2 Cure characteristics of EPDM/SBR nanocomposites filled with tCNTs

Sample code	Cure characteristics				
	Torques		Times		Cure rate index (CRI = $100/t_{90}$) (t_{90} , min ⁻¹)
	Min. torque (M_L , dN m)	Max. torque (M_H , dN m)	Torque difference ($\Delta M = M_H - M_L$) (dN m)	Scorch time (t_{s2} , min.)	Optimum cure time (t_{90} , min.)
tCNTs ₀	1.01	6.3	5.29	4.18	8.5
tCNTs ₁	1.05	6.44	5.39	2.77	6.91
tCNTs ₂	1.09	6.54	5.45	2.67	6.72
tCNTs ₃	1.13	6.66	5.53	2.54	6.48
tCNTs ₄	1.16	6.74	5.58	2.4	6.22
tCNTs ₅	1.19	6.98	5.79	2.11	5.82
tCNTs ₆	1.2	7.02	5.82	2.09	5.75
tCNTs ₇	1.21	7.11	5.9	2.05	5.7
tCNTs ₈	1.22	7.18	5.96	2.02	5.65

deformation, reflected in the higher M_L values. Additionally, the functionalization of CNTs with TESPT establishes chemical bonds between the filler and the matrix, further strengthening the reinforcement and limiting the free movement of polymer chains during processing, thereby contributing to the increase in M_L .

In Table 2, the M_H of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. As the concentration of tCNTs increases, the M_H of the EPDM/SBR nanocomposites also increases. With a greater amount of tCNTs, these nanotubes act as effective fillers, enhancing the stiffness and rigidity of the rubber matrix by creating a strong filler-polymer network [69]. The tCNTs improve dispersion and promote robust interfacial bonding between the nanotubes and the EPDM/SBR matrix, which reduces the mobility of the polymer chains and enhances load transfer during deformation. These enhanced interactions result in greater resistance to applied stress, leading to higher M_H values, which reflect the improved strength and rigidity of the nanocomposites. Additionally, the nanotube network can restrict movement under shear forces, contributing to the overall increase in torque.

In Table 2, the ΔM of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. With higher tCNT content, the ΔM of the tCNT-reinforced EPDM/SBR nanocomposites shows an upward trend. ΔM , which represents the difference between M_H and M_L , reflects the CLD and stiffness of the material after curing. As the tCNT content increases, the nanotubes form a stronger, more interconnected filler network within the EPDM/SBR blend matrix. This results in enhanced stress transfer, increased stiffness, and restricted mobility of the polymer chains. tCNTs promote better dispersion and chemical bonding with the polymer, further reinforcing the material. Consequently, the M_H (which indicates material strength) increases, while the M_L (which relates to the viscosity of the uncured material) shows a smaller change, leading to a higher ΔM . The increase in CLD and overall rigidity of the nanocomposites with higher tCNT content also contributes to this upward trend in ΔM .

In Table 2, the t_{s2} of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. An increase in tCNT content results in a shorter t_{s2} for the tCNT-reinforced EPDM/SBR nanocomposites. tCNTs possess high thermal conductivity, which facilitates faster heat dissipation within the nanocomposites [70]. This enhanced thermal conductivity leads to a more uniform temperature distribution during processing, accelerating the onset of crosslinking reactions and thus reducing t_{s2} [70]. tCNTs can also affect the curing kinetics, as the functional groups on the tCNTs may engage with the curing agents in the EPDM/SBR blend. This interaction could modify the curing

dynamics and shorten the time needed for the material to initiate crosslinking, resulting in a decreased t_{s2} . Moreover, enhanced dispersion of tCNTs within the polymer matrix minimizes localized overheating and promotes more uniform processing conditions, which further aids in reducing t_{s2} .

In Table 2, the t_{90} of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. With higher concentrations of tCNTs, the t_{90} in the tCNT-reinforced EPDM/SBR nanocomposites tends to decrease. tCNTs possess high thermal conductivity, which helps distribute heat more efficiently throughout the composite material, promoting faster and more uniform curing of the polymer matrix [71]. As a result, the overall curing process accelerates, leading to a decrease in t_{90} . tCNTs also promote a more even distribution throughout the EPDM/SBR matrix. This better dispersion prevents localized overheating and ensures consistent curing conditions across the material, reducing the time required to reach the optimum cure state. Furthermore, the functional groups on the tCNTs may interact with curing agents or reactants in the EPDM/SBR blend, potentially accelerating crosslinking reactions. This catalytic effect accelerates the completion of the curing process. The strengthened interactions between the tCNTs and the polymer matrix likely contribute to more efficient curing by reducing viscosity and promoting faster crosslinking, leading to a shorter t_{90} time.

In Table 2, the CRI of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. Increasing the tCNT content results in a higher CRI in the EPDM/SBR nanocomposites. tCNTs possess high thermal conductivity, enhancing heat distribution within the composite during the curing process [72]. This efficient heat transfer accelerates the curing reactions, leading to a higher CRI, which reflects the speed at which the material reaches its optimal state of crosslinking. The functional groups on tCNTs improve the interaction between the CNTs and the EPDM/SBR matrix, facilitating a more effective curing process by enhancing the integration of curing agents and accelerating crosslinking reactions. tCNTs also help achieve uniform dispersion within the polymer matrix, reducing localized overheating and ensuring even distribution of curing agents throughout the material. As a result, the curing process becomes faster and more consistent, contributing to a higher CRI. Additionally, tCNTs may have a catalytic effect on the curing reactions, promoting faster crosslinking. This can increase the rate of cure as the curing agents react more quickly with the polymer matrix. The presence of tCNTs can also reduce the viscosity of the polymer blend, enabling more efficient mixing and faster curing. Lower viscosity facilitates quicker heat transfer and reaction kinetics, thereby increasing the CRI.

Figure 1 illustrates the stress–strain behavior of EPDM/SBR nanocomposites. The stress increased progressively with strain, similar to the response seen in the EPDM/SBR blend. As the tCNT content in the EPDM/SBR nanocomposites increased, stress at a given strain also rose, peaking at 5 phr of tCNT before decreasing. From this stress–strain curve, three key parameters were obtained: tensile strength, stress at 100% elongation, and elongation at break. The tensile properties of the EPDM/SBR–tCNT nanocomposites are shown in Figs. 2 and 3.

In Fig. 2, the tensile strength (TS) of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. The TS increases as the tCNT content rises to 5 phr, after which it decreases with additional tCNT incorporation. Functionalizing CNTs with TESPT enhances their dispersion in the EPDM/SBR matrix, minimizing agglomeration and fostering a more uniform distribution of CNTs throughout the matrix. This enhanced dispersion improves the efficiency of load transfer between the CNTs and the EPDM/SBR. TESPT also creates chemical bonds between the CNTs and the rubber, increasing interfacial adhesion, which facilitates more effective load transfer, thereby improving

the TS of the composite. The high aspect ratio, modulus, and strength of CNTs reinforce the blend matrix, contributing to the enhanced mechanical properties, including TS. Up to 5 phr, the amount of CNTs is sufficient to enhance these properties without negatively impacting the material [73]. At higher concentrations, CNTs may start to agglomerate due to insufficient dispersion and increased filler–filler interactions [74]. This agglomeration can result in localized weaknesses within the rubber matrix, which subsequently reduces TS [75]. Beyond a certain point, the added CNTs may disrupt the rubber matrix structure more than they reinforce it. The excessive amount of filler can interfere with the rubber's elasticity and reduce the effectiveness of stress transfer, resulting in decreased TS. High concentrations of CNTs can increase the viscosity of the rubber compound, making it more difficult to process and resulting in poorer mixing and dispersion. This can lead to defects and inconsistencies in the material, affecting its mechanical properties. When compared to the standard EPDM/SBR blend vulcanizates, the TS of the EPDM/SBR nanocomposites demonstrated an increase of 177%. The addition of TESPT-modified CNTs into the EPDM/SBR matrix contributes to

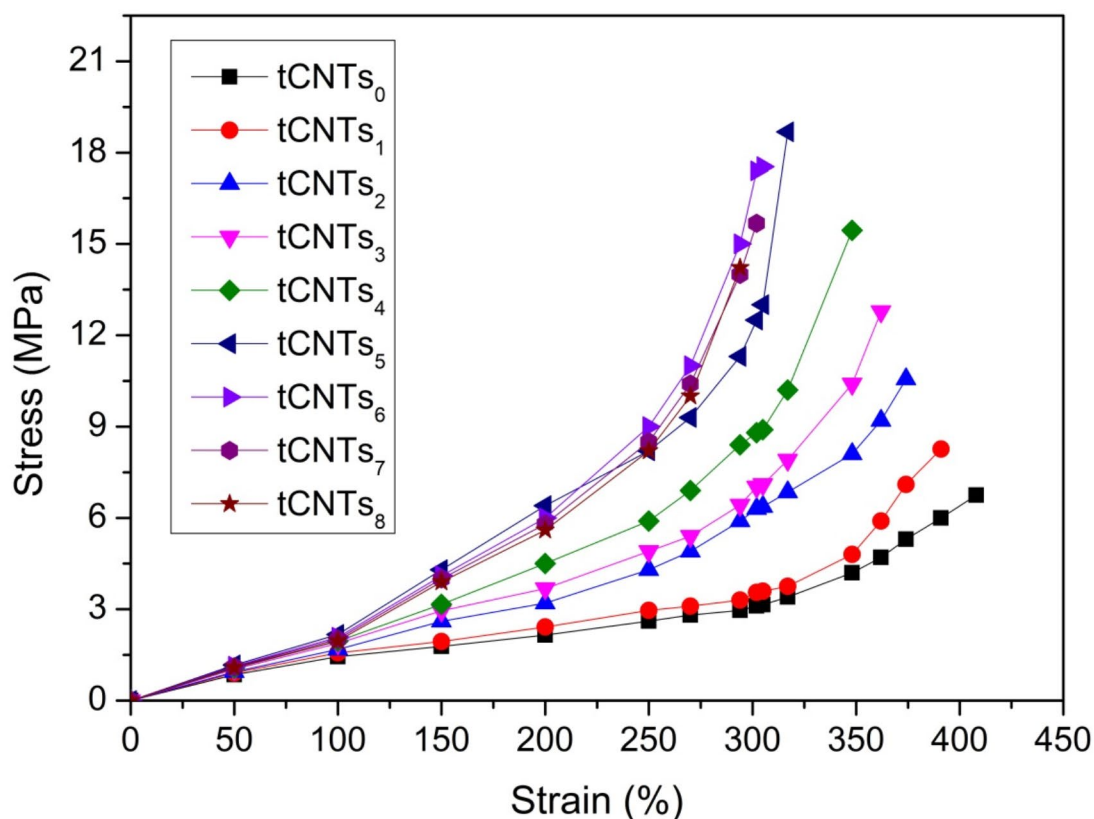


Fig. 1 Stress-strain curve of EPDM/SBR nanocomposites with tCNTs addition

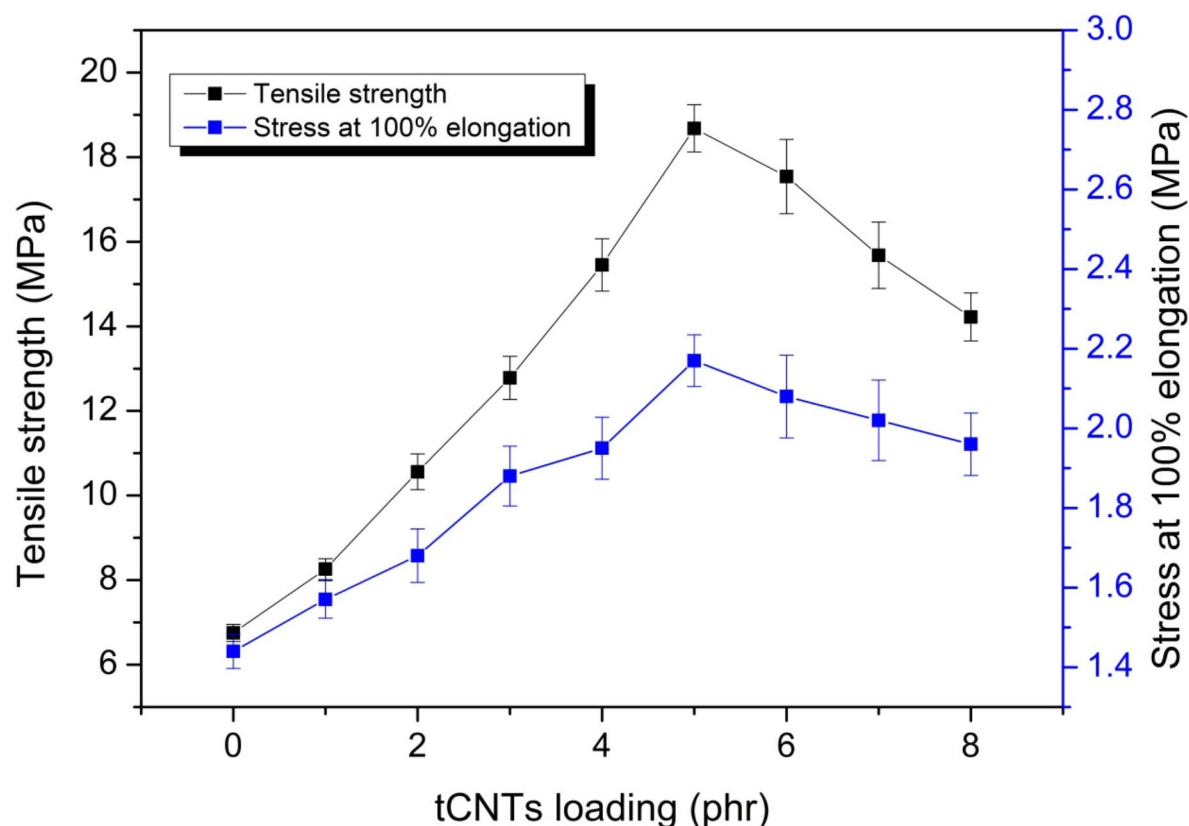


Fig. 2 Tensile strength (TS) and stress at 100% elongation (M100) in EPDM/SBR nanocomposites with tCNT addition

an increase in CLD. As a coupling agent, TESPT establishes chemical bonds between the CNTs and the rubber matrix. This additional cross-linking can enhance the overall network structure of the rubber, improving its mechanical properties, including TS. The increased CLD helps distribute the applied load more effectively throughout the material, thereby increasing its strength. TESPT modifies the CNTs, enhancing their compatibility with the rubber blend matrix. This modification enhances the interactions between the CNTs and the rubber, facilitating more effective load transfer from the EPDM/SBR matrix to the CNTs. Stronger filler-matrix interactions mean that the CNTs can better reinforce the rubber network, contributing to increased TS. The improved CLD ensures that these interactions are more effective. The addition of TESPT-modified CNTs can lead to a more tightly cross-linked network within the rubber matrix. This denser network improves the material's structural integrity and its ability to resist deformation under tensile stress. Consequently, the tensile strength of the EPDM/SBR nanocomposites is considerably greater than that of the base EPDM/SBR blend, which exhibits a lower cross-link density and weaker interactions between the filler and

matrix. A higher CLD leads to more uniform stress distribution throughout the material [76]. This uniform distribution minimizes localized stress concentrations and reduces the likelihood of failure under tensile loads. The improved stress distribution contributes to the enhanced TS observed in the nanocomposites. With increased CLD, the rubber matrix becomes more resistant to deformation and stretching. This enhanced resistance allows the material to retain its strength and structural integrity under tensile stress, resulting in the observed improvement in TS.

In Fig. 2, the stress at 100% elongation (M100) of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. The M100 of the EPDM/SBR nanocomposites rises with the incorporation of tCNTs up to 5 phr, but begins to decline with further increases in tCNT content. This increase in M100 (also known as modulus) with the addition of tCNTs up to 5 phr can be attributed to improved reinforcement from the well-dispersed CNTs. When tCNTs are included, they serve as effective reinforcing fillers due to their high aspect ratio, robust interfacial bonding with the EPDM/SBR blend matrix (aided by TESPT functionalization), and their ability to limit the

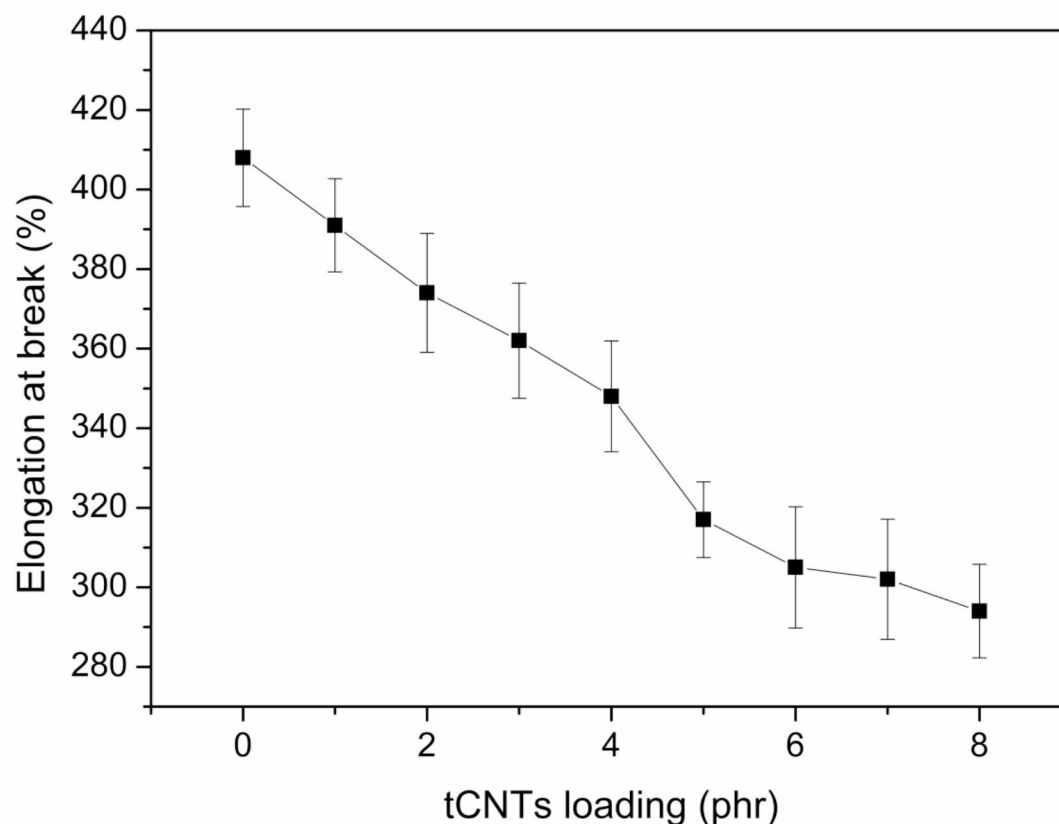


Fig. 3 Elongation at Break (EB) in EPDM/SBR nanocomposites with tCNT addition

mobility of the rubber chains. This creates a stiffer network capable of withstanding higher stress at a given elongation, resulting in an increase in modulus. However, when the tCNT content surpasses 5 phr, the M100 begins to decrease. At higher loadings, tCNTs tend to aggregate due to van der Waals forces, leading to a non-uniform dispersion in the EPDM/SBR blend matrix. This agglomeration diminishes the effective surface area of CNTs interacting with the rubber matrix, diminishing the reinforcing effect. With excessive tCNTs, the optimal balance between filler and crosslinked rubber network may be disrupted. Higher filler content can interfere with the vulcanization process, reducing the efficiency of crosslink formation, which negatively impacts the mechanical properties. Overloading of tCNTs can make the nanocomposite more brittle, as the increased filler content reduces the matrix flexibility. This leads to a less flexible material that fails to distribute stress efficiently, resulting in a drop in M100 [73]. Compared to the original base blend vulcanizates, the M100 of the EPDM/SBR nanocomposites increases by 51%. The incorporation of TESPT-modified CNTs promotes the development of additional crosslinks within the rubber matrix. TESPT acts as a

coupling agent, forming chemical bonds between the CNTs and the rubber (EPDM/SBR) matrix during the vulcanization process. This leads to an increase in CLD, creating a more rigid and tightly interconnected network structure within the rubber. Consequently, the composite necessitates greater force to achieve 100% elongation, indicating a rise in stress at this level of elongation. The improved cross-link density from the interaction between CNTs and the EPDM/SBR matrix restricts the movement of the polymer chains. This limited chain mobility leads to a stiffer network that is less capable of deforming under applied stress, resulting in an increased modulus. The more crosslinked structure offers greater resistance to elongation, contributing to the 51% increase in M100.

In Fig. 3, the elongation at break (EB) of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. As the tCNT content increases, the EB of the EPDM/SBR nanocomposites exhibits a downward trend. The addition of tCNTs improves the CLD of the composite. While a higher cross-link density enhances the mechanical strength of the material, it also restricts the movement of the polymer chains. Consequently, the material becomes more

rigid and less able to elongate, resulting in a decrease in elongation at break. The robust interfacial bonding between the tCNTs and the EPDM/SBR matrix improves stress transfer but concurrently limits the flexibility of the polymer chains. This stiffer network diminishes the material's capacity to withstand significant deformations prior to failure. At higher tCNT concentrations, there is a tendency for CNTs to agglomerate. This leads to filler-filler interactions that create stress concentrations within the matrix, causing early failure during tensile testing [77]. This further contributes to a reduction in EB as the material becomes more brittle. The EB of the EPDM/SBR nanocomposites decreased by 28% compared to the base EPDM/SBR blend vulcanizates. This reduction in EB is due to the heightened stiffness and decreased flexibility of the rubber matrix resulting from the addition of tCNTs. The inclusion of tCNTs improves the interactions between the EPDM/SBR and the tCNTs, creating a stiffer network within the matrix. As a result, the material's capacity to deform under stress is reduced, leading to the decline in EB. Furthermore, the increased cross-link density from the tCNTs further restricts the movement

of the polymer chains, contributing to the overall decrease in elongation.

In Fig. 4, the tear strength (TAS) of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. As the content of tCNTs rises, the tensile adhesive strength (TAS) of the EPDM/SBR nanocomposites also increases. This enhancement is largely attributed to the improved interfacial bonding between the rubber matrix and the filler, resulting in more effective load transfer during stress [78–80]. The chemical bonding between the tCNTs and the rubber chains, facilitated by the TESPT (SCA), results in a more cohesive and stronger composite structure [81–82]. Moreover, the even distribution of tCNTs throughout the rubber matrix plays a key role in forming a robust reinforcing network that aids in preventing crack propagation during tear testing. The presence of tCNTs acts as a barrier to crack growth by dissipating the applied stress more effectively across the material. This enhanced filler-rubber interaction and improved dispersion of tCNTs leads to increased resistance to tearing forces, thereby increasing the TAS of the composite. Lastly, the tCNTs contribute to the overall stiffness and strength of the composite by increasing

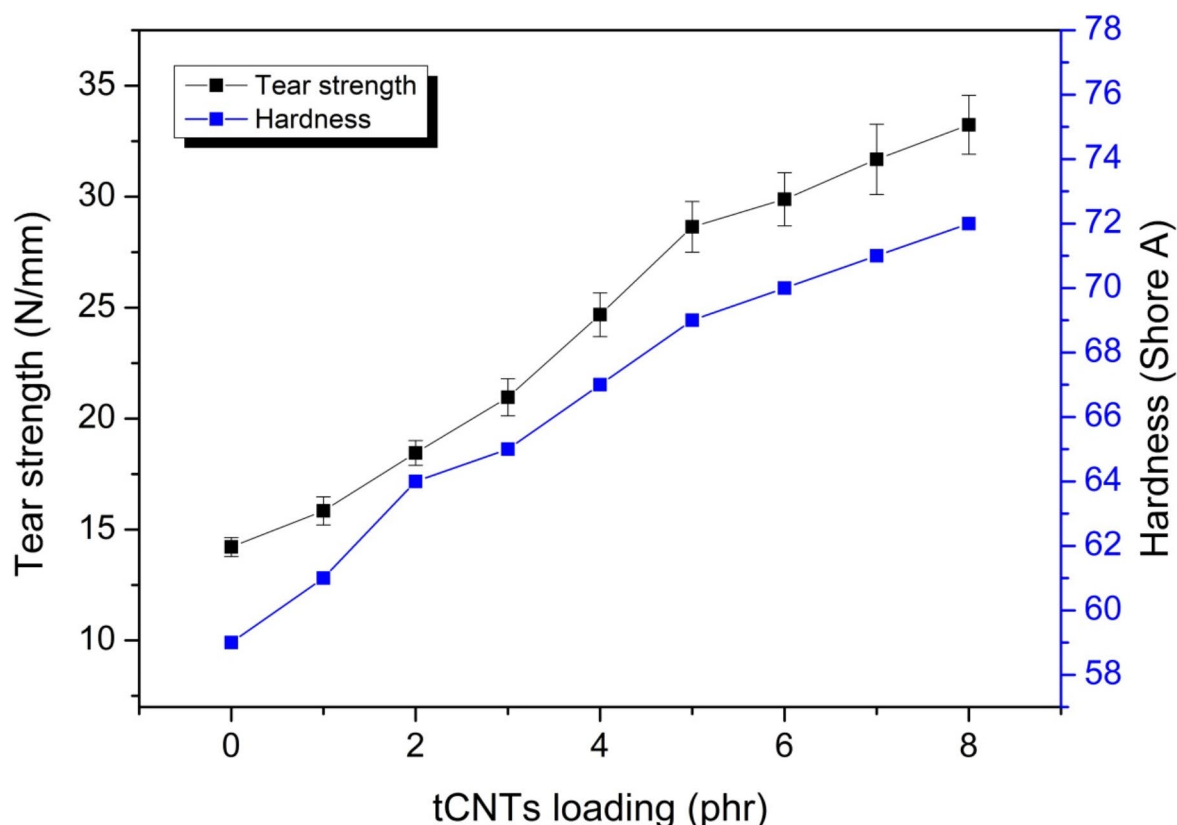


Fig. 4 Tear strength (TAS) and hardness in EPDM/SBR nanocomposites with tCNT addition

the CLD, which further enhances its ability to withstand higher stresses without failure. These combined effects explain the observed improvement in TAS with the addition of tCNTs. The TAS of the EPDM/SBR nanocomposites shows a 134% improvement over the standard EPDM/SBR blend vulcanizates.

In Fig. 4, the hardness of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. The incorporation of tCNTs leads to an increase in the hardness of these composites. Adding tCNTs to the EPDM/SBR matrix enhances the composite's stiffness. CNTs are known for their high mechanical strength and stiffness. When incorporated into the rubber matrix, they strengthen the material, enhancing its rigidity and resulting in increased hardness. The increased filler content enhances the composite's overall stiffness, as the CNTs form a more rigid network structure within the blend (EPDM/SBR) matrix [83]. The use of TESPT as a coupling agent facilitates the formation of extra chemical bonds between the CNTs and the EPDM/SBR matrix during vulcanization, resulting in an increased CLD of the EPDM/SBR matrix. This higher CLD creates a more closely linked polymer network, contributing to enhanced

hardness. The presence of crosslinks reduces the flexibility of the polymer chains, thereby producing a harder composite. Additionally, the efficient dispersion of tCNTs within the rubber matrix enhances the interaction between the filler and the matrix. By ensuring proper dispersion, a consistent distribution of CNTs throughout the matrix is achieved, which maximizes their reinforcing potential. This uniform reinforcement contributes to an overall increase in hardness. Moreover, the chemical bonding facilitated by TESPT strengthens the interaction between the CNTs and the rubber matrix, resulting in a more stable and rigid composite structure that ultimately enhances hardness.

In Fig. 5, the rebound resilience (RR) of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. The reduction in RR with increasing tCNT content in EPDM/SBR nanocomposites can be explained by the heightened stiffness and rigidity provided by the tCNTs. When tCNTs are incorporated into the EPDM/SBR matrix, they form a reinforcing network that restricts the movement of the polymer chains. This results in a more rigid composite structure, contributing to the material's increased hardness. The increase in hardness is due to the

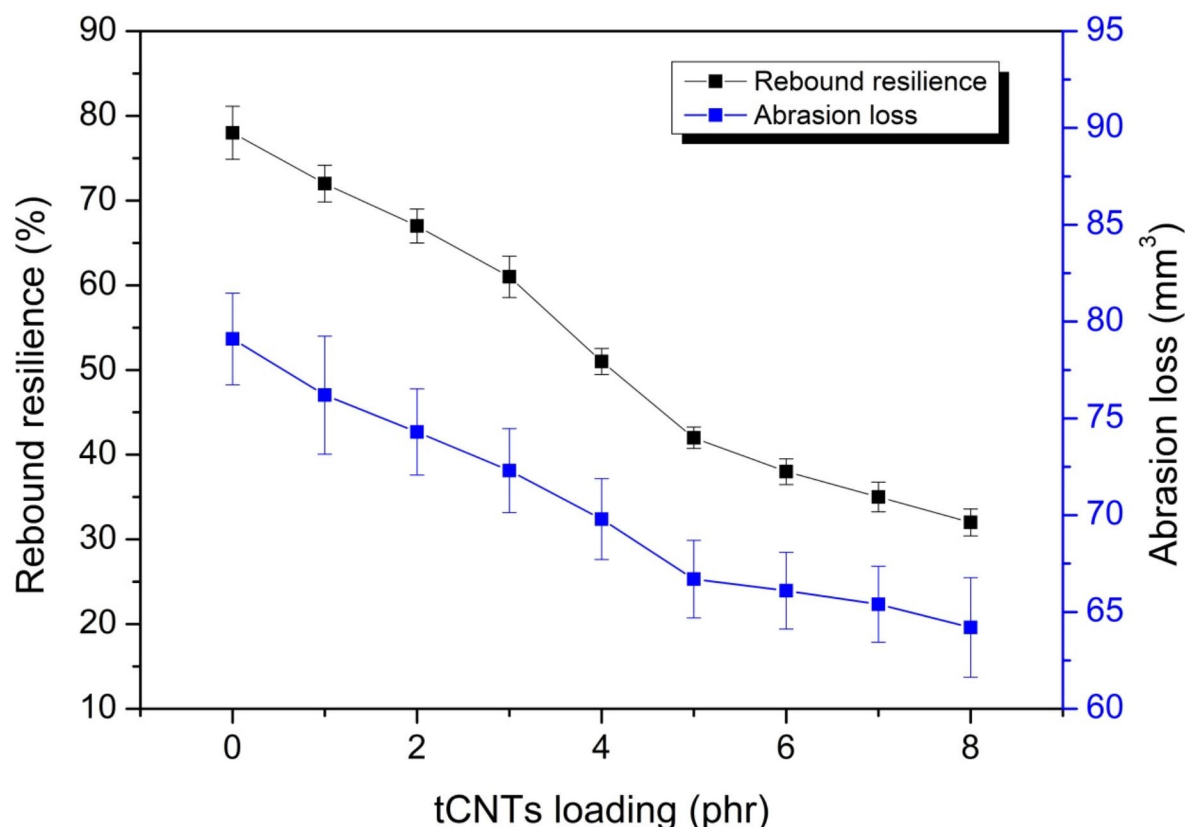


Fig. 5 Rebound resilience (RR) and abrasion loss in EPDM/SBR nanocomposites with tCNT addition

enhanced cross-linking and the reinforcing characteristics of the tCNTs, which improve the composite's stiffness and mechanical strength. However, this increased rigidity limits the elastomer's ability to deform and recover quickly after impact, which directly affects the material's resilience. RR is a measure of how well a material can recover from deformation, and as the matrix becomes stiffer due to tCNTs, the material's ability to rebound decreases. Thus, the observed inverse relationship between hardness and RR results from the trade-off between rigidity and elasticity in the nanocomposites. As more tCNTs are added, the material becomes progressively harder and less able to revert to its original shape after deformation, leading to a decline in resilience. When compared to the original EPDM/SBR blend vulcanizates, the RR of the EPDM/SBR nanocomposites shows a significant decrease of 59%.

In Fig. 5, the abrasion loss of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. As the content of tCNTs increases, the abrasion loss in the EPDM/SBR nanocomposites declines, signifying improved abrasion resistance (AR). This reduction in abrasion loss is mainly due to the enhanced reinforcement offered by the tCNTs. The effective interaction between the CNTs and the rubber matrix, aided by the SCA, promotes better dispersion of the CNTs throughout the matrix. As a result, this leads to a more even distribution of stress during mechanical loading, thereby reducing material wear during abrasion. Additionally, the high TS and M100 of CNTs contribute to the nanocomposites' increased resistance to deformation and surface damage under frictional forces. The tCNTs form a network structure within the rubber matrix, which strengthens the material, leading to improved AR. As the CLD of the nanocomposites also increases with tCNT content, the rigidity and structural integrity of the material are further enhanced, reducing the likelihood of surface wear and abrasion loss. However, beyond an optimal level of CNT loading, the benefits may plateau or diminish due to potential agglomeration of CNTs, which could disrupt the uniformity of the matrix and lead to localized stress concentrations. Therefore, the increase in AR is most effective up to a certain tCNT content.

In Fig. 6, the compression set (CS) of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. The base compounds exhibited the lowest CS value, whereas the CS of the EPDM/SBR nanocomposites increases with a higher tCNT content. The addition of tCNTs enhances crosslinking in the EPDM/SBR matrix by improving the interaction between the rubber chains and the CNTs. Although an increased CLD usually enhances mechanical properties such as TS and tear resistance, it may also diminish the elasticity and recovery of the rubber, resulting in a higher CS. This happens as the material becomes more rigid

and less capable of returning to its original shape after compression. tCNTs serve as reinforcing fillers in the EPDM/SBR matrix, enhancing the stiffness of the composite while diminishing its flexibility. As the stiffness increases, the material becomes less able to recover fully after prolonged deformation, which contributes to a higher CS. This occurrence is typical in highly filled composites, where the fillers restrict the movement of the polymer chains. The addition of tCNTs increases the rigidity of the nanocomposite, making the polymer network less resilient to long-term compressive stress. As a result, the ability of the composite to recover from deformation is compromised, leading to higher residual deformation, which is observed as a higher CS. A lower CS is generally preferred for better performance in applications [68].

In Fig. 6, the CLD of the nanocomposites made from EPDM/SBR and reinforced with tCNTs is illustrated. The rise in CLD with the incorporation of tCNTs up to 5 phr can be linked to better dispersion and strong interfacial bonding between the modified CNTs and the EPDM/SBR matrix. The TESPT acts as a coupling agent, enhancing the chemical bonding between the rubber chains and CNTs, which promotes a higher degree of crosslinking. This leads to a more tightly interconnected network, increasing the CLD [83]. However, when the tCNT content exceeds 5 phr, the CLD starts to decline, likely due to the agglomeration of CNTs. An excess of CNTs tends to form clusters, which disrupts their uniform distribution within the EPDM/SBR matrix. These agglomerates reduce the effective surface area of the CNTs, diminishing their reinforcing effect and weakening the interfacial bonding between the EPDM/SBR blend matrix and the filler. Consequently, this leads to a decrease in the overall CLD. Additionally, higher CNT content may also hinder the vulcanization process, further contributing to the decline in CLD at higher loadings.

The tensile fracture surface of the EPDM/SBR rubber blend nanocomposites is shown in Fig. 7, captured using FESEM. Figure 7a displays the EPDM/SBR blend, revealing a homogeneous mixing of the two rubbers without noticeable phase separation. This uniformity indicates good compatibility between EPDM and SBR, likely due to their similar polarities or molecular structures, which facilitate effective blending. Figure 7b depicts EPDM/SBR nanocomposites reinforced with 5 phr of tCNTs, where the tCNTs are uniformly dispersed throughout the rubber matrix. This even distribution suggests optimal dispersion at this concentration and effective interaction between the tCNTs and the EPDM/SBR matrix, likely enhanced by the functionalization of the tCNTs, which improves bonding with the matrix. In contrast, Fig. 7c illustrates EPDM/SBR nanocomposites reinforced with 7 phr of tCNTs, showing noticeable agglomeration within the matrix. This agglomeration indicates that

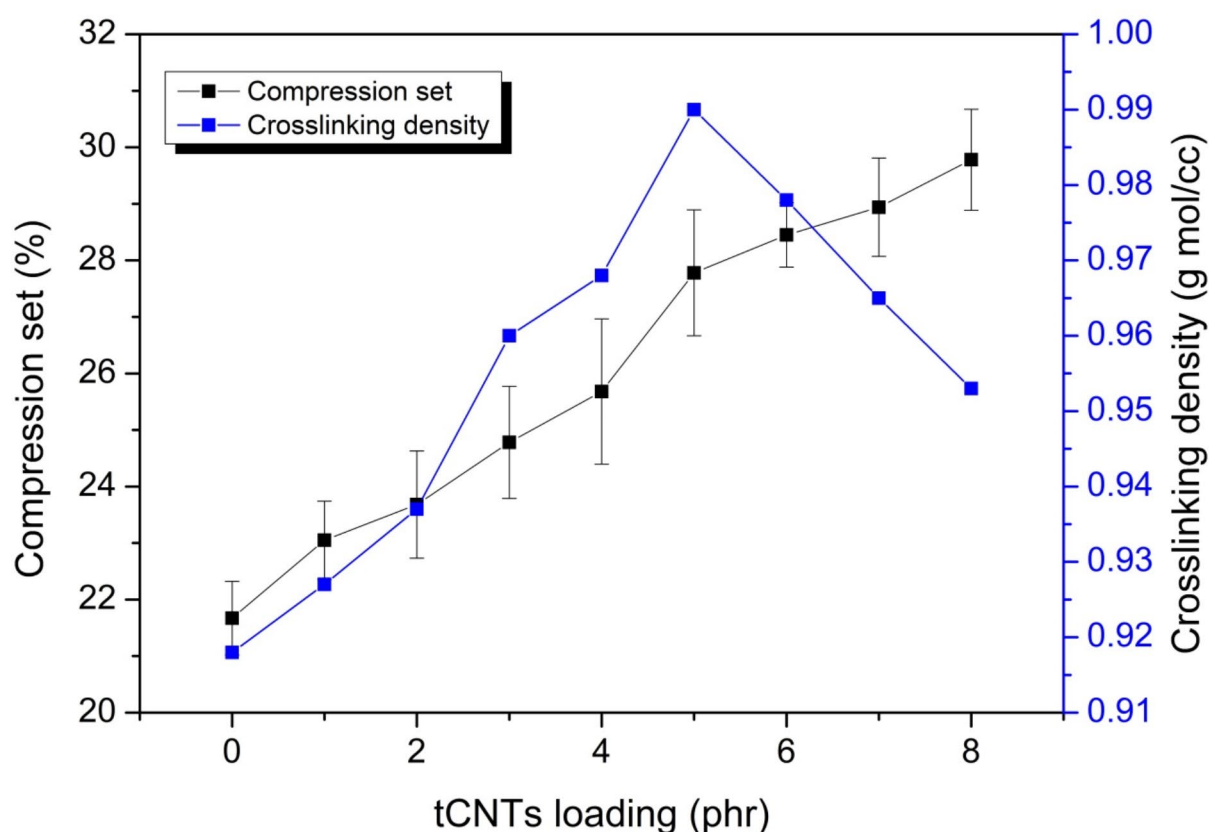


Fig. 6 Compression set (CS) and crosslinking density (CLD) in EPDM/SBR nanocomposites with tCNT addition

the 7 phr concentration may have exceeded the percolation threshold, leading to cluster formation. At higher concentrations, the tendency of tCNTs to aggregate due to van der Waals forces results in poor dispersion and agglomeration, which can adversely affect the overall properties of the nanocomposite.

Presented in Table 3 are the mole percent uptake (MPU) values for the EPDM/SBR nanocomposites in a range of solvents. The base rubber blend without any tCNT reinforcement shows the highest MPU. Upon the introduction of tCNTs into the matrix (EPDM/SBR blend), the MPU decreases. As tCNTs are added to the EPDM/SBR matrix, they occupy space within the rubber, leading to a reduction in the free volume available for solvent diffusion. This reduction in free volume means that there are fewer spaces within the matrix for the solvent to penetrate, which can result in lower MPU values. The incorporation of tCNTs can enhance the CLD in the rubber matrix by facilitating better interactions between the functionalized CNTs and the EPDM/SBR blend. Enhanced crosslinking creates a more rigid network structure, which can impede the diffusion of solvents into the rubber matrix, thus reducing the MPU.

tCNTs can interact with the EPDM/SBR matrix through chemical bonding or physical interactions. These interactions may alter the matrix's ability to absorb solvents. The functionalization of CNTs (e.g., with TESPT) improves compatibility with the rubber matrix, potentially leading to stronger filler-rubber interactions that can reduce the solvent uptake. At elevated concentrations, tCNTs can cluster together within the rubber matrix. These clusters may form areas that impede solvent penetration, leading to a further decrease in the MPU. The aggregation of CNTs disrupts the homogeneous distribution of the filler, leading to uneven solvent absorption. The reinforcement effect of tCNTs can make the rubber matrix more rigid and less permeable to solvents. The increased rigidity and reduced matrix flexibility can contribute to a lower MPU, as the rubber is less able to accommodate and absorb the solvent.

Among aromatic solvents, benzene exhibits the greatest absorption, followed by toluene, xylene, and mesitylene. Benzene, being the smallest of the aromatic solvents listed, has the smallest molecular size and the highest surface area-to-volume ratio. This allows it to more readily penetrate and interact with the rubber matrix, resulting in greater

Fig. 7 FESEM images of **a** EPDM/SBR blends, **b** EPDM/SBR reinforced with 5 phr of tCNTs, and **c** EPDM/SBR reinforced with 7 phr of tCNTs

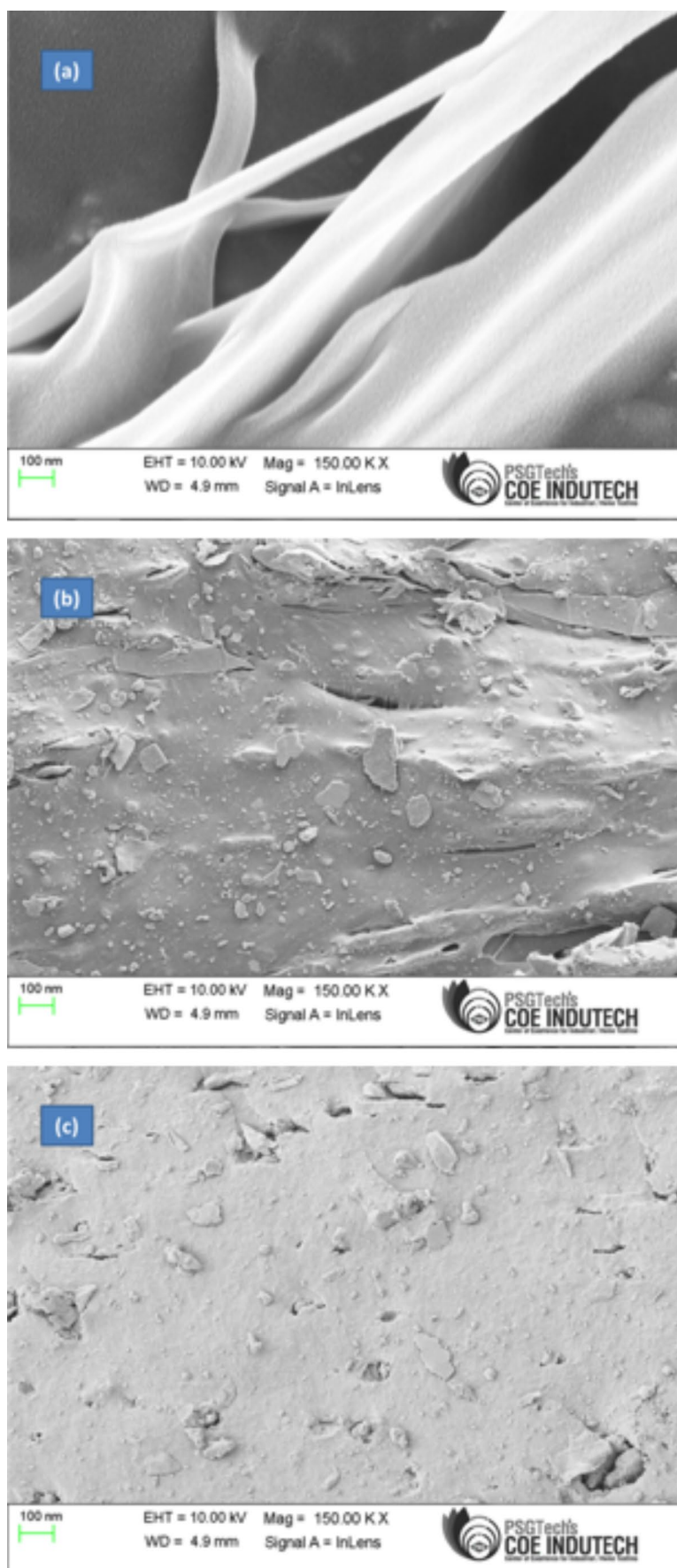


Table 3 MPU values of EPDM/SBR-tCNTs nanocomposites in different solvents

Sample code	Mole percent uptake (mol%) at 30°C									
	Aromatic				Aliphatic				Chlorinated	
	Benzene	Toluene	Xylene	Mesitylene	n-pentane	n-hexane	n-heptane	n-octane	Dichloromethane	Chloroform
tCNTs ₀	3.95	3.57	3.46	3.03	2.42	2.34	2.28	2.24	5.42	4.56
tCNTs ₁	3.9	3.51	3.34	2.91	2.34	2.26	2.2	2.15	5.27	4.35
tCNTs ₂	3.87	3.47	3.29	2.85	2.31	2.22	2.16	2.11	5.23	4.31
tCNTs ₃	3.82	3.41	3.24	2.81	2.27	2.18	2.12	2.06	5.17	4.24
tCNTs ₄	3.79	3.38	3.21	2.76	2.24	2.15	2.08	2.02	5.12	4.21
tCNTs ₅	3.74	3.32	3.15	2.68	2.18	2.09	2.03	1.97	5.05	4.15
tCNTs ₆	3.76	3.34	3.17	2.72	2.19	2.11	2.07	1.99	5.08	4.17
tCNTs ₇	3.79	3.36	3.19	2.74	2.21	2.14	2.1	2.03	5.11	4.2
tCNTs ₈	3.81	3.38	3.22	2.77	2.23	2.16	2.12	2.05	5.15	4.24

absorption. In contrast, larger molecules like toluene, xylene, and mesitylene have lower surface area-to-volume ratios and more steric hindrance, which can limit their ability to diffuse into the matrix. Benzene, with its relatively nonpolar nature, has a strong tendency to interact with other nonpolar substances in the rubber matrix. This strong affinity facilitates its absorption. As the aromatic rings become substituted (as in toluene, xylene, and mesitylene), the polarity of the molecules can increase, leading to less favorable interactions with the nonpolar rubber matrix. Consequently, the absorption decreases. As the aromatic rings of the solvents become substituted with additional groups (e.g., methyl groups in toluene, xylene, and mesitylene), the size and steric effects of these groups can hinder the solvent molecules' ability to fit into and interact with the porous structure of the rubber. This leads to reduced absorption compared to benzene, which lacks such substituents. Benzene is more volatile and has a higher diffusion rate compared to the larger, more complex aromatic solvents. This higher volatility and ease of diffusion contribute to benzene's greater absorption capacity. Toluene, xylene, and mesitylene have lower diffusion rates due to their larger molecular sizes and reduced volatility, leading to lower absorption. The intermolecular forces between the aromatic solvents themselves also play a role. Benzene, being a simple aromatic hydrocarbon, does not experience strong intermolecular forces compared to substituted aromatics. This lack of strong intermolecular interactions allows benzene to be absorbed more readily into the rubber matrix compared to its substituted counterparts.

Aliphatic solvents are absorbed in the following order: n-pentane, n-hexane, n-heptane, and n-octane, with n-pentane demonstrating the highest absorption and n-octane the lowest. As the carbon chain length extends from n-pentane (5 carbon atoms) to n-octane (8 carbon atoms), the size of the solvent molecules increases. Larger molecules, such as n-octane, possess a more complex molecular structure, making it harder for them to penetrate the rubber matrix compared to smaller molecules like n-pentane. Thus, shorter chain aliphatic hydrocarbons, like n-pentane, are absorbed more readily due to their smaller size, allowing them to diffuse through the EPDM/SBR matrix more effortlessly. Aliphatic hydrocarbons primarily interact with the EPDM/SBR matrix through van der Waals forces. The strength of these interactions tends to increase with the size of the molecule. However, larger aliphatic hydrocarbons also face steric hindrance, which limits their ability to diffuse and be absorbed by the matrix. This balance results in lower absorption rates for larger molecules such as n-heptane and n-octane. The solubility parameter, which indicates a solvent's compatibility with the polymer, decreases with increasing solvent chain length. Solvents with a higher solubility parameter, like n-pentane, exhibit greater compatibility with the polymer

matrix, resulting in higher absorption rates. However, as the chain length progresses from n-pentane to n-octane, the solubility parameter declines, leading to a reduced level of absorption. Shorter aliphatic molecules like n-pentane have higher diffusivity, meaning they can penetrate the polymer matrix faster than longer chains such as n-octane. This results in higher uptake of shorter chain solvents. Conversely, the larger, more complex structure of n-octane impedes its diffusion rate, leading to lower absorption.

The order of absorption for chlorinated solvents is dichloromethane having the greatest uptake, followed by chloroform, and carbon tetrachloride showing the least absorption. Dichloromethane has the smallest molecular size and the highest polarity compared to the other two chlorinated solvents. Its smaller size and higher polarity enhance its ability to interact with and be absorbed by the rubber matrix. The increased polarity of dichloromethane allows it to form stronger interactions with the polar groups in the polymer matrix, leading to greater absorption. Dichloromethane's lower molecular weight and higher dipole moment contribute to its stronger interactions with the rubber matrix compared to chloroform and carbon tetrachloride. These interactions facilitate better diffusion and absorption into the polymer network. Chloroform, with a slightly larger molecular size and lower polarity compared to dichloromethane, exhibits intermediate absorption properties. Its molecular interactions with the rubber matrix are less favorable than those of dichloromethane but better than those of carbon tetrachloride [84–85]. Carbon tetrachloride has the largest molecular size and is the least polar among the three solvents. Its non-polar nature results in weaker interactions with the polar groups in the EPDM/SBR matrix, leading to lower absorption compared to dichloromethane and chloroform. The variations in absorption can also be linked to the solubility of the solvents within the EPDM/SBR matrix. Dichloromethane's high solubility facilitates greater uptake, whereas carbon tetrachloride's lower solubility results in reduced absorption. The solubility parameter of each solvent affects its ability to diffuse and be retained within the polymer matrix.

4 Conclusions

In this study, the impact of various types and concentrations of tCNTs on the properties of EPDM/SBR nanocomposites was investigated.

1. The addition of tCNTs to EPDM/SBR nanocomposites improves material properties by increasing both the M_L and M_H torques, indicating enhanced stiffness and stress transfer. The ΔM rises with higher tCNT content due to increased CLD. Additionally, tCNTs reduce t_{s2} and t_{90} , and increase the CRI, reflecting faster and more efficient curing processes. Overall, tCNTs enhance the performance of EPDM/SBR nanocomposites by optimizing their rheological and curing properties.
2. The TS of the EPDM/SBR nanocomposites exhibited a significant increase of up to 150% compared to the base EPDM/SBR blend, primarily resulting from improved CLD with optimal tCNT levels (up to 5 phr). However, a further rise in tCNT content resulted in a reduction in TS, likely due to the agglomeration of the nanotubes, which negatively impacts mechanical performance. Similarly, the M100 increased with the addition of tCNTs up to 5 phr but decreased at higher concentrations due to the same agglomeration effects.
3. The EB diminished as the tCNT content increased. This reduction is attributed to the rigidifying effect of tCNTs, which reduces the polymer matrix's flexibility and elongation capability.
4. The hardness of the EPDM/SBR nanocomposites rose with the addition of tCNTs, suggesting that the material gained stiffness due to enhanced reinforcement. In contrast, RR decreased as tCNT content increased. This negative correlation between hardness and RR stems from the heightened rigidity introduced by the tCNTs, which reduces the material's capacity to return to its original shape after deformation.
5. The abrasion loss of the EPDM/SBR nanocomposites diminished as the tCNT content increased, indicating enhanced abrasion resistance. This improvement is probably attributed to the stronger interactions between the tCNTs and the rubber matrix, which bolster the material's resistance to wear.
6. The CLD of the EPDM/SBR nanocomposites rose with increasing tCNT content up to 5 phr, resulting in improved mechanical properties. However, a decline in CLD at higher tCNT concentrations was observed due to agglomeration.
7. The CS of the nanocomposites increased with tCNT content, with a lower CS generally indicating better performance.
8. FESEM analysis of the tensile fracture surfaces revealed a uniform distribution of tCNTs in the matrix at lower concentrations, indicating effective dispersion. At higher concentrations, agglomeration of tCNTs was observed, which may negatively impact the properties of the nanocomposites.
9. The absorption behavior of the EPDM/SBR nanocomposites varied with solvent type. Aromatic solvents were absorbed in the order: benzene > toluene > xylene > mesitylene, reflecting the decreasing polarity and molecular size. Aliphatic solvents were

absorbed in the order: n-pentane > n-hexane > n-heptane > n-octane, correlating with decreasing molecular size and increasing hydrocarbon chain length. Chlorinated solvents were absorbed in the order: dichloromethane > chloroform > carbon tetrachloride, which reflects the decreasing polarity and molecular size of the solvents.

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Author Contributions Dr. S. Vishvanathperumal was instrumental in the experimental phase of this study, managing the execution of various tests and experiments. He meticulously prepared all the figures presented in the manuscript, ensuring clarity and accuracy. Furthermore, Dr. S. Vishvanathperumal took the lead in writing the main manuscript text, providing detailed accounts of the experimental procedures, results, and interpretations. Mr. K.N. Ramu provided valuable guidance and oversight throughout the project. In addition to supervising the experimental work, he addressed issues related to plagiarism in the main manuscript. His expertise in academic integrity ensured that the manuscript complied with the highest ethical standards. Dr. V. Navaneethakrishnan and Mr. K. Parthasarathy made important contributions to the manuscript by helping to identify and rectify potential instances of plagiarism. Their thorough review and keen attention to detail were crucial in preserving the integrity of the research findings. All authors, including Dr. S. Vishvanathperumal, Mr. K.N. Ramu, Dr. V. Navaneethakrishnan, and Mr. K. Parthasarathy, actively participated in the comprehensive review of the main manuscript. Their collaborative efforts ensured the manuscript's accuracy, coherence, and overall quality before submission.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests. The authors affirm that they have no conflicts of interest to report, stating that none of their financial, professional, or personal interests might influence the study's conclusions or interpretations.

Consent to Participate There were neither human nor animal participants in this scientific project. The study design and methods were purposefully developed to avoid using human subjects or animals at any stage of the research process, and ethical issues were carefully taken into account.

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References

1. K. Manimaran, S. Vishvanathperumal, 2024, April. On the potential of clay with respect to conventional fillers (CB and SiO₂) for ENR compatibilized NR/EPDM vulcanizates. In AIP Conference Proceedings (Vol. 3037, No. 1). AIP Publishing
2. J. Srinivas, M.S. Jagatheeshwaran, S. Vishvanathperumal, G. Anand, The effect of nanosilica on mechanical and swelling resistance properties of ternary rubber (NR/SBR/NBR) blends nanocomposites with and without bis (triethoxysilylpropyl) tetrasulfane. *Silicon*. **16**(4), 1669–1688 (2024)
3. K. Parthasarathy, S. Vishvanathperumal, T. Balasubramanian, R.R. Anirudh, 2024, April. Enhancement of the mechanical properties of EPDM/NBR rubbers using nanosilica for seal applications. In AIP Conference Proceedings (Vol. 3037, No. 1). AIP Publishing
4. K. Karthikeyan, A. Damodaran, S. Vishvanathperumal, 2024. Effect of nano-silica surface-capped by bis [3-(triethoxysilyl) propyl] tetrasulfide on the cure behaviors, mechanical properties, swelling resistance and microstructure of styrene-butadiene rubber/acrylonitrile-butadiene rubber nanocomposites. *Journal of Polymer Research*, 31(3), p.79
5. S. Vishvanathperumal, V. Navaneethakrishnan, S. Gopalakannan, The effect of Nanoclay and hybrid filler on curing characteristics, mechanical properties and swelling resistance of ethylene-vinyl acetate/styrene butadiene rubber blend composite. *J. Adv. Microsc. Res.* **13**(4), 469–476 (2018)
6. S. Vishvanathperumal, S. Gopalakannan, Swelling properties, compression set behavior and abrasion resistance of ethylene-propylene-diene rubber/styrene butadiene rubber blend nanocomposites. *Polym. Korea*. **41**(3), 433–442 (2017)
7. S.P. Kumar, G. Prabhakaran, S. Vishvanathperumal, Influence of modified Nanosilica on the performance of NR/EPDM blends: cure characteristics, mechanical properties and Swelling Resistance. *J. Inorg. Organomet. Polym. Mater.* **34**, 3420–3442 (2024)
8. G. Sundaravadivel, S.R. Venkataraman, S. Vishvanathperumal, V. Navaneethakrishnan, Influence of APTES modified HNTs on properties of NR/EPDM nanocomposites. *Silicon*. **15**, 6715–6727 (2023)
9. S. Vishvanathperumal, J.V. Roy, G. Anand, K.N. Ramu, S. Praveenkumar, An investigation on the effect of the surface modifications and HNTs loading on the cure behaviours, abrasion resistance, mechanical and morphological properties of NR/EPDM nanocomposites. *Silicon*. **16**(5), 2267–2284 (2024)
10. P.S. Ganeche, P. Balasubramanian, S. Vishvanathperumal, Halloysite Nanotubes (HNTs)-Filled Ethylene-propylene-diene Monomer/Styrene-Butadiene Rubber (EPDM/SBR) composites: mechanical, swelling, and Morphological properties. *Silicon*. **14**, 6611–6620 (2022)
11. S. Vishvanathperumal, G. Anand, Effect of nanosilica on the mechanical properties, compression set, morphology, abrasion and swelling resistance of sulphur cured EPDM/SBR composites. *Silicon*. **14**(7), 3523–3534 (2022)
12. M.S. Ravi Theja, N. Kilari, S. Vishvanathperumal, V. Navaneethakrishnan, Modeling tensile modulus of nanoclay-filled ethylene-propylene-diene monomer/styrene-butadiene rubber using composite theories. *J. Rubber Res.* **24**(5), 847–856 (2021)
13. S. Vishvanathperumal, G. Anand, Effect of Nanosilica and Cross-linking System on the Mechanical Properties and Swelling Resistance of EPDM/SBR nanocomposites with and without TESPT. *Silicon*. **13**(10), 3473–3497 (2021)
14. S. Vishvanathperumal, G. Anand, Effect of nanoclay/nanosilica on the mechanical properties, abrasion and swelling resistance of EPDM/SBR composites. *Silicon*. **12**(8), 1925–1941 (2020)

15. S. Vishvanathperumal, V. Navaneethakrishnan, G. Anand, S. Gopalakannan, Evaluation of crosslink density using material constants of ethylene-propylene-diene monomer/styrene-butadiene rubber with different nanoclay loading: finite element analysis-simulation and experimental. *Adv. Sci. Eng. Med.* **12**(5), 632–642 (2020)
16. T. Xu, Z. Jia, Y. Luo, D. Jia, Z. Peng, Interfacial interaction between the epoxidized natural rubber and silica in natural rubber/silica composites. *Appl. Surf. Sci.* **328**, 306–313 (2015)
17. R. Sengupta, S. Chakraborty, S.A. Bandyopadhyay, S. Dasgupta, R. Mukhopadhyay, K. Auddy, A.S. Deuri, A short review on rubber/clay nanocomposites with emphasis on mechanical properties. *Polym. Eng. Sci.* **47**(11), 1956–1974 (2007)
18. Q. Fang, B. Song, T.T. Tee, L.T. Sin, D. Hui, S.T. Bee, Investigation of dynamic characteristics of nano-size calcium carbonate added in natural rubber vulcanizate. *Compos. Part. B: Eng.* **60**, 561–567 (2014)
19. E. Farida, N. Bukit, E.M. Ginting, B.F. Bukit, 2019. The effect of carbon black composition in natural rubber compound. *Case Studies in Thermal Engineering*, 16, p.100566
20. E.S. Kim, T.H. Lee, E.J. Kim, J.S. Yoon, Surface modification of carbon fiber and the mechanical properties of the silicone rubber/carbon fiber composites. *J. Appl. Polym. Sci.* **126**(S2), E410–E418 (2012)
21. D. Yang, J.R. Tao, Y. Yang, Q.M. He, Y.X. Weng, B. Fei, M. Wang, 2022. Effect interfacial size and multiple interface on electromagnetic shielding of silicon rubber/carbon nanotube composites with mixing segregated particles. *Composite Structures*, 292, p.115668
22. Z. Xie, Y. Cai, Y. Zhan, Y. Meng, Y. Li, Q. Xie, H. Xia, 2022. Thermal insulating rubber foams embedded with segregated carbon nanotube networks for electromagnetic shielding applications. *Chemical Engineering Journal*, 435, p.135118
23. J. Shao, H. Zhu, B. Zhao, S.I. Haruna, G. Xue, W. Jiang, K. Wu, J. Yang, 2022. Combined effect of recycled tire rubber and carbon nanotubes on the mechanical properties and microstructure of concrete. *Construction and Building Materials*, 322, p.126493
24. M.N. Alam, V. Kumar, D.J. Lee, J. Choi, 2023. Synergistically toughened silicone rubber nanocomposites using carbon nanotubes and molybdenum disulfide for stretchable strain sensors. *Composites Part B: Engineering*, 259, p.110759
25. D. Liu, Q.Q. Kong, H. Jia, L.J. Xie, J. Chen, Z. Tao, Z. Wang, D. Jiang, C.M. Chen, Dual-functional 3D multi-wall carbon nanotubes/graphene/silicone rubber elastomer: thermal management and electromagnetic interference shielding. *Carbon*. **183**, 216–224 (2021)
26. H. Yang, L. Yuan, X. Yao, Z. Zheng, D. Fang, 2020. Monotonic strain sensing behavior of self-assembled carbon nanotubes/graphene silicone rubber composites under cyclic loading. *Composites Science and Technology*, 200, p.108474
27. X. Guo, T. Liu, Y. Tang, W. Li, L. Liu, D. Wang, Y. Zhang, T. Zhang, X. Zhu, Y. Guan, X. Li, Bioinspired Low Hysteresis Flexible pressure Sensor using nanocomposites of Multiwalled Carbon nanotubes, Silicone Rubber, and Carbon Nanofiber for Human–Computer Interaction. *ACS Appl. Nano Mater.* **7**(13), 15626–15639 (2024)
28. B. Yang, S. Wang, Z. Song, L. Liu, H. Li, Y. Li, Molecular dynamics study on the reinforcing effect of incorporation of graphene/carbon nanotubes on the mechanical properties of swelling rubber. *Polym. Test.* **102**, 107337 (2021)
29. L. Százdí, A. Pozsgay, B. Pukánszky, Factors and processes influencing the reinforcing effect of layered silicates in polymer nanocomposites. *Eur. Polymer J.* **43**(2), 345–359 (2007)
30. C.L. Wu, M.Q. Zhang, M.Z. Rong, K. Friedrich, Tensile performance improvement of low nanoparticles filled-polypropylene composites. *Compos. Sci. Technol.* **62**(10–11), 1327–1340 (2002)
31. R.J. Young, S.J. Eichhorn, Deformation mechanisms in polymer fibres and nanocomposites. *Polymer*. **48**(1), 2–18 (2007)
32. S. Iijima, Helical microtubules of graphitic carbon. *Nature*. **354**(6348), 56–58 (1991)
33. E.W. Wong, P.E. Sheehan, C.M. Lieber, Nanobeam mechanics: elasticity, strength, and toughness of nanorods and nanotubes. *Science*. **277**(5334), 1971–1975 (1997)
34. M.J. Treacy, T.W. Ebbesen, J.M. Gibson, Exceptionally high Young's modulus observed for individual carbon nanotubes. *Nature*. **381**(6584), 678–680 (1996)
35. R.S. Ruoff, D.C. Lorents, Mechanical and thermal properties of carbon nanotubes. *Carbon*. **33**(7), 925–930 (1995)
36. V. Aravinth, V. Navaneethakrishnan, S. Vishvanathperumal, G. Gurumoorthi, Effect of modified nanographene oxide (mGO)/carbon nanotubes (CNTs) hybrid filler on the cure, mechanical and swelling properties of silicone rubber composites. *J. Inorg. Organomet. Polym. Mater.* **34**(1), 282–301 (2023)
37. S. Endo, Y. Terada, Y. Kato, I. Nakai, Chemical speciation of arsenic-accumulating mineral in a sedimentary iron deposit by synchrotron radiation multiple X-ray analytical techniques. *Environ. Sci. Technol.* **42**(19), 7152–7158 (2008)
38. M. Endo, Y.A. Kim, Y. Fukai, T. Hayashi, M. Terrones, H. Terrones, M.S. Dresselhaus, Comparison study of semi-crystalline and highly crystalline multiwalled carbon nanotubes. *Appl. Phys. Lett.* **79**(10), 1531–1533 (2001)
39. M.S. Dresselhaus, G. Dresselhaus, P.C. Eklund, *Science of Fullerenes and Carbon Nanotubes: Their Properties and Applications* (Elsevier, 1996)
40. A.M. Shanmugharaj, J.H. Bae, K.Y. Lee, W.H. Noh, S.H. Lee, S.H. Ryu, Physical and chemical characteristics of multiwalled carbon nanotubes functionalized with aminosilane and its influence on the properties of natural rubber composites. *Compos. Sci. Technol.* **67**(9), 1813–1822 (2007)
41. K. Kueseng, K.I. Jacob, Natural rubber nanocomposites with SiC nanoparticles and carbon nanotubes. *Eur. Polymer J.* **42**(1), 220–227 (2006)
42. M.A. López-Manchado, J. Biagiotti, L. Valentini, J.M. Kenny, Dynamic mechanical and Raman spectroscopy studies on interaction between single-walled carbon nanotubes and natural rubber. *J. Appl. Polym. Sci.* **92**(5), 3394–3400 (2004)
43. S. Joly, G. Garnaud, R. Ollitrault, L. Bokobza, J.E. Mark, Organically modified layered silicates as reinforcing fillers for natural rubber. *Chem. Mater.* **14**(10), 4202–4208 (2002)
44. S. Varghese, J. Karger-Kocsis, Natural rubber-based nanocomposites by latex compounding with layered silicates. *Polymer*. **44**(17), 4921–4927 (2003)
45. Y.A. Kim, T. Hayashi, M. Endo, Y. Gotoh, N. Wada, J. Seiyama, Fabrication of aligned carbon nanotube-filled rubber composite. *Scripta Mater.* **54**(1), 31–35 (2006)
46. A. Fakhru'l-Razi, M.A. Atieh, N. Girun, T.G. Chuah, M. El-Sadig, D.R.A. Biak, Effect of multi-wall carbon nanotubes on the mechanical properties of natural rubber. *Compos. Struct.* **75**(1–4), 496–500 (2006)
47. N. Yan, J.K. Wu, Y.H. Zhan, H.S. Xia, Carbon nanotubes/carbon black synergistic reinforced natural rubber composites. *Plast., Rubber Compos.* **38**(7), 290–296 (2009)
48. Y. Nakaramontri, C. Kummerlöwe, C. Nakason, N. Vennemann, The effect of surface functionalization of carbon nanotubes on properties of natural rubber/carbon nanotube composites. *Polym. Compos.* **36**(11), 2113–2122 (2015)
49. S. Dhanasekar, S. Baskar, S. Vishvanathperumal, Halloysite Nanotubes Effect on Cure and Mechanical properties of EPDM/NBR nanocomposites. *J. Inorg. Organomet. Polym. Mater.* **33**, 3208–3220 (2023)
50. K. Ragupathy, G. Prabakaran, N. Pragadish, S. Vishvanathperumal, Effect of silica nanoparticles and modified

- silica nanoparticles on the mechanical and Swelling properties of EPDM/SBR Blend nanocomposites. *Silicon*. **15**, 6033–6046 (2023)
51. Y. Yang, S.W. Ng, D. Chen, J. Chang, D. Wang, J. Shang, Q. Huang, Y. Deng, Z. Zheng, 2019. Freestanding lamellar porous carbon stacks for low-temperature-foldable supercapacitors. *Small*, **15**(48), p.1902071
 52. R.K. Das, K. Ragupathy, T.S. Kumar, S. Vishvanathperumal, Effect of Halloysite Nanotubes (HNTs) on Mechanical properties of EPDM/NBR blend-nanocomposites. *Polym. Korea*. **47**(2), 221–232 (2023)
 53. R. Sundar, S.K. Mohan, S. Vishvanathperumal, Effect of Surface modified Halloysite Nanotubes (mHNTs) on the Mechanical properties and Swelling Resistance of EPDM/NBR nanocomposites. *Polym. Korea*. **46**(6), 728–743 (2022)
 54. G. Anand, S. Vishvanathperumal, Properties of SBR/NR Blend: the effects of Carbon Black/Silica (CB/SiO₂) hybrid filler and silane Coupling Agent. *Silicon*. **14**(14), 9051–9060 (2022)
 55. S. Vishvanathperumal, S. Gopalakannan, *Effects of the Nanoclay and Crosslinking Systems on the Mechanical Properties of ethylene-propylene-diene monomer/styrene Butadiene Rubber Blends Nanocomposite*, vol. 11 (*Silicon*, 2019), pp. 117–135
 56. S. Vishvanathperumal, S. Gopalakannan, *Reinforcement of Ethylene Vinyl Acetate with Carbon black/silica Hybrid Filler Composites*, vol. 852 (*Applied Mechanics and Materials*, 2016), pp. 16–22
 57. K. Senthilvel, S. Vishvanathperumal, B. Prabu, J. Baruch, L. Studies on the morphology, cure characteristics and mechanical properties of acrylonitrile butadiene rubber with hybrid filler (carbon black/silica) composite. *Polym. Polym. Compos.* **24**(7), 473–480 (2016)
 58. M. Karthikeyan, E.G. Sundaram, S. Vishvanathperumal, Development of halloysite nanotubes reinforced chlorinated ethylene propylene diene monomer/chlorinated acrylonitrile butadiene rubber blends. *J. Polym. Res.* **31**(6), 1–22 (2024)
 59. K. Govindan, S. Ramabalan, S. Vishvanathperumal, Enhancement of Mechanical Performance and Swelling Resistance in Silicone Rubber through reinforcement with γ -Methacryloxypropyltrimethoxysilane-modified Halloysite Nanotubes. *J. Inorg. Organomet. Polym. Mater.* **34**(5), 2242–2259 (2024)
 60. V. Sivaramakrishnan, S.R. Venkataraman, S. Vishvanathperumal, V. Navaneethakrishnan, Improved mechanical performance and swelling resistance of ethylene propylene diene monomer/styrene butadiene rubber nanocomposites through the incorporation of graphene oxide as a reinforcing filler. *J. Polym. Res.* **31**(10), 313 (2024)
 61. A. Arunkumar, S.R. Venkataraman, S. Vishvanathperumal, V. Navaneethakrishnan, *Comparative Analysis of Cure Behaviors, Mechanical Properties, and Swelling Resistance in EPDM/SBR Composites with HNTs, APTES-Modified HNTs, and RH-Modified HNTs*, vol. 16 (*Silicon*, 2024), pp. 5691–5714
 62. A. Arunkumar, D. Srinivasan, S. Vishvanathperumal, V. Navaneethakrishnan, Effect of HNTs and modified HNTs nanotubes on the mechanical properties and swelling resistance of EPDM/SBR rubber blend nanocomposites. *Silicon*. **15**(17), 7647–7667 (2023)
 63. G. Sundaravadivel, K. Parthasarathy, S. Vishvanathperumal, V. Navaneethakrishnan, Effect of complex of resorcinol and hexamethylenetetramine modified halloysite nanotubes (RH-HNTs) on the mechanical and swelling characteristics of NR/EPDM nanocomposites. *J. Polym. Res.* **30**, 383 (2023)
 64. K. Govindan, S. Ramabalan, S. Vishvanathperumal, S. Chockalingam, 2023. Influence of halloysite nanotubes on mechanical and swelling properties of silicone rubber compound. *Journal of Polymer Research*, **30**(8), p.310
 65. P.C. Prakash, G. Gurumoorthi, V. Navaneethakrishnan, S. Vishvanathperumal, Effect of nanographene oxide on the mechanical properties of EPDM/SBR nano-composites. *Polym. Korea*. **47**(4), 427–439 (2023)
 66. V. Aravinth, G. Gurumoorthi, S. Vishvanathperumal, V. Navaneethakrishnan, Effect of modified Nanographene Oxide on the mechanical and Swelling properties of Silicone Rubber nanocomposites. *Polym. Korea*. **47**(3), 288–302 (2023)
 67. S. Dhanasekar, S. Baskar, S. Vishvanathperumal, 2023. Cure characteristics, compression set, swelling behaviors, abrasion resistance and mechanical properties of nanoclay (Cloisite 15A, Cloisite 20A and Cloisite 30B) filler filled EPDM/NBR blend system. *Journal of Polymer Research*, **30**(10), p.375
 68. H. Hu, L. Zhao, J. Liu, Y. Liu, J. Cheng, J. Luo, Y. Liang, Y. Tao, X. Wang, J. Zhao, Enhanced dispersion of carbon nanotube in silicone rubber assisted by graphene. *Polymer*. **53**(15), 3378–3385 (2012)
 69. J. Abraham, H.J. Maria, S.C. George, N. Kalarikkal, S. Thomas, Transport characteristics of organic solvents through carbon nanotube filled styrene butadiene rubber nanocomposites: the influence of rubber–filler interaction, the degree of reinforcement and morphology. *Phys. Chem. Chem. Phys.* **17**(17), 11217–11228 (2015)
 70. G. Sui, W.H. Zhong, X.P. Yang, Y.H. Yu, S.H. Zhao, Preparation and properties of natural rubber composites reinforced with pre-treated carbon nanotubes. *Polym. Adv. Technol.* **19**(11), 1543–1549 (2008)
 71. Y. Nakaramontri, C. Nakason, C. Kummerlöwe, N. Vennemann, Enhancement of electrical conductivity and filler dispersion of carbon nanotube filled natural rubber composites by latex mixing and in situ silanization. *Rubber Chem. Technol.* **89**(2), 272–291 (2016)
 72. D. Ponnammma, K.T. Varughese, M.A.A. Al-Maadeed, S. Thomas, Curing enhancement and network effects in multi-walled carbon nanotube-filled vulcanized natural rubber: evidence for solvent sensing. *Polym. Int.* **66**(6), 931–938 (2017)
 73. H.H. Le, S. Abhijeet, S. Ilisch, J. Klehm, S. Henning, M. Beiner, S.S. Sarkawi, W. Dierkes, A. Das, D. Fischer, K.W. Stöckelhuber, The role of linked phospholipids in the rubber–filler interaction in carbon nanotube (CNT) filled natural rubber (NR) composites. *Polymer*. **55**(18), 4738–4747 (2014)
 74. L. Bokobza, Elastomer nanocomposites: effect of filler–matrix and filler–filler interactions. *Polymers*. **15**(13), 2900 (2023)
 75. F. Danafar, M. Kalantari, A review of natural rubber nanocomposites based on carbon nanotubes. *J. Rubber Res.* **21**, 293–310 (2018)
 76. K. Senthil nathan, S. Krishnamohan, V. Navaneetha krishnan, S. Vishvanath Perumal, *Study on the Effect of Halloysite Nanotubes on the Mechanical Properties and Swelling Resistance of Ethylene-Propylene Diene Monomer (EPDM)/Nitrile Butadiene Rubber (NBR) Blend Nanocomposites* (*Materials Research Ex*, 2024)
 77. V. Kumar, M.N. Alam, A. Manikkavel, M. Song, D.J. Lee, S.S. Park, 2021. Silicone rubber composites reinforced by carbon nanofillers and their hybrids for various applications: A review. *Polymers*, **13**(14), p.2322
 78. C. Nah, J.Y. Lim, B.H. Cho, C.K. Hong, A.N. Gent, Reinforcing rubber with carbon nanotubes. *J. Appl. Polym. Sci.* **118**(3), 1574–1581 (2010)
 79. Y. Nakaramontri, C. Nakason, C. Kummerlöwe, N. Vennemann, Influence of modified natural rubber on properties of natural rubber–carbon nanotube composites. *Rubber Chem. Technol.* **88**(2), 199–218 (2015)
 80. X. Zhou, Y. Zhu, J. Liang, S. Yu, New fabrication and mechanical properties of styrene-butadiene rubber/carbon nanotubes nanocomposite. *J. Mater. Sci. Technol.* **26**(12), 1127–1132 (2010)

81. G. Li, C.S. Morlor, C. Leung, H. Wang, 2021. Mechanical properties and fractal analysis of cement mortar incorporating styrene-butadiene rubber latex and carboxylated MWCNTs. *Construction and Building Materials*, 309, p.125175
82. L. Tzounis, S. Debnath, S. Rooj, D. Fischer, E. Mäder, A. Das, M. Stamm, G. Heinrich, *High Performance Natural Rubber Composites with a Hierarchical Reinforcement Structure of Carbon Nanotube Modified Natural Fibers*, vol. 58 (*Materials & Design*, 2014), pp. 1–11
83. X. Kuang, G. Liu, X. Dong, D. Wang, Enhancement of mechanical and self-healing performance in multiwall carbon nanotube/rubber composites via Diels–Alder bonding. *Macromol. Mater. Eng.* **301**(5), 535–541 (2016)
84. P.C. Prakash, D. Srinivasan, V. Navaneethakrishnan, S. Vishvanathperumal, Effect of modified nanographene oxide loading on the swelling and compression set behavior of EPDM/SBR nano-composites. *J. Inorg. Organomet. Polym. Mater.* **34**(2), 593–610 (2023)
85. S. Dhanasekar, M.R. Theja, S. Baskar, S. Vishvanathperumal, Effects of Nanosilica on the Mechanical properties and Swelling Resistance of EPDM/NBR nanocomposites. *Polym. Korea.* **47**(5), 613–627 (2023)

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